

TRANSMISSION LINES & WAVEGUIDES (ECA15)

Name	A.K.LIKITH CHARAN	Register No.	192412305
Date	07/10/2026	Experiment No.	1

Experiment No.: 1

Title: Investigating the significance of Characteristic Impedance (Z_0) and Propagation Constant (γ) of a Transmission Line using Resistance, Capacitance, Inductance and Conductance.

AIM

To investigate the significance of the characteristic impedance (Z_0) and propagation constant (γ) of a transmission line by varying its primary constants and observing the resulting attenuation constant (α) and phase constant (β).

APPARATUS / SOFTWARE REQUIRED

- SCILAB Version 6.1.1

THEORY / CALCULATIONS USED

The report follows the equations given in the laboratory manual:

Characteristic impedance: $Z_0 = \sqrt{[(R + j\omega L)/(G + j\omega C)]}$

Propagation constant: $\gamma = \sqrt{[(R + j\omega L)(G + j\omega C)]} = \alpha + j\beta$

where α is the attenuation constant in neper/km and β is the phase constant in radians/km.

PROCEDURE

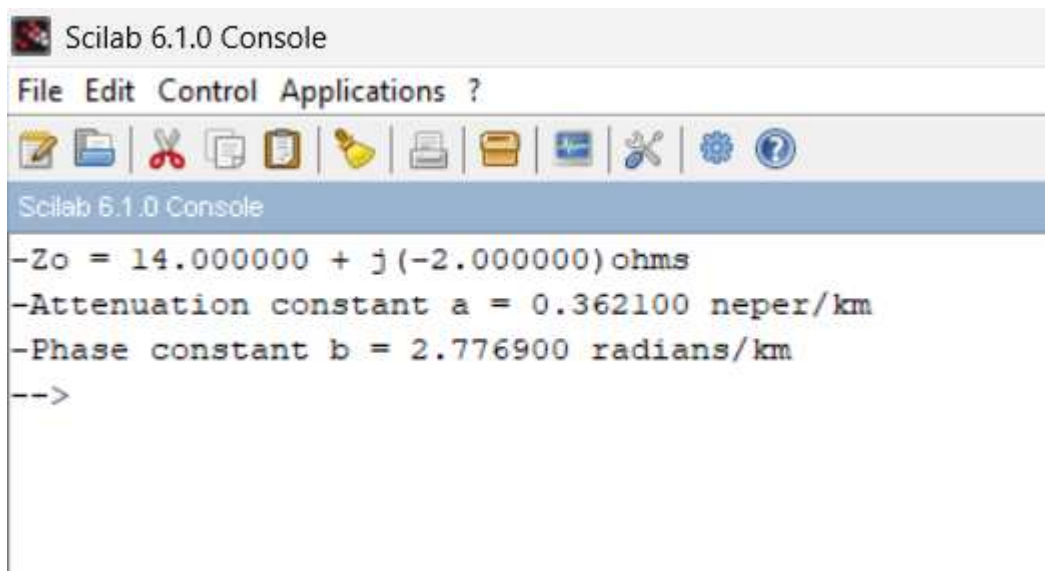
- Open SCILAB software.
- Open File → New and create a new program.
- Enter the required values of R, L, G, C and frequency.
- Enter the SCILAB program and execute it.
- Record Z_0 , α and β from the console output.
- For Test Cases 2 and 3, change only the intended parameter(s) and repeat the calculation.
- Compare the simulated/console values with the theoretical values calculated using the same equations.

TEST CASE 1

CODE

```
clear; clc;
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);
w=2*pi*f;
Z=R+(i*w*L);
Y=G+(i*w*C);
Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);
```

OUTPUT SCREENSHOT



```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console
-Zo = 14.000000 + j(-2.000000) ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km
-->
```

TABLE

Run	R	L	G	C	f	Z_0 (Ω)	α (Np/km)	β (rad/km)
Basic run	10 Ω	3 mH	4 μ S	16 μ F	2 kHz	14 - j2 Ω	0.3621	2.7769

RESULT – TEST CASE 1

The characteristic impedance and propagation constant were obtained using SCILAB. For the supplied program configuration, the console output shows approximately $Z_0 = 14 - j2 \Omega$, $\alpha = 0.3621$ Np/km and $\beta \approx 2.7765$ rad/km. The result demonstrates that the secondary transmission-line parameters depend directly on the primary constants.

TEST CASE 2 – VARYING RESISTANCE AND INDUCTANCE

The manual asks for $f = 2.5$ kHz, resistance varied from 10 Ω to 50 Ω in 10 Ω steps, and inductance varied from 3.5 mH to 11.5 mH in 2.5 mH steps, while the remaining constants are held fixed. □filecite□turn0file0□L156-L169□

CODE VARYING RESISTANCE

```
clear;

clc;

// Fixed parameters

f = 2.5*10^3;

L = 3.5*10^-3;

G = 4*10^-6;

C = 8*10^-6;

w = 2*pi*f;

// Resistance values: 10 to 50 ohms

R_values = [10 20 30 40 50];

disp("TEST CASE 2A: VARYING RESISTANCE");

for k = 1:5

    R = R_values(k);

    Z = R + %i*w*L;

    Y = G + %i*w*C;

    Zo = sqrt(Z/Y);

    gamma = sqrt(Z*Y);

    disp("Resistance R (ohms):");

    disp(R);

    disp("Characteristic Impedance Zo (ohms):");

    disp(Zo);

    disp("Attenuation Constant alpha (neper/km):");

    disp(real(gamma));

    disp("Phase Constant beta (radians/km):");

    disp(imag(gamma));

    disp("-----");

end
```

RESISTANCE VARIATION

R (Ω)	Z_0 (Ω)	α (Np/km)	β (rad/km)
10	21 -2j	0.2381	2.6392
20	21 -4j	0.4707	2.6702
30	22 -6j	0.6935	2.7184
40	22 -7j	0.9042	2.7796
50	23 -9j	1.1023	2.8502

INDUCTANCE VARIATION

L (mH)	Z_0 (Ω)	α (Np/km)	β (rad/km)
3.5	21 -2j	0.2381	2.6392
6.0	27 -1j	0.1824	3.4463
8.5	33 -1j	0.1534	4.0990
11.0	37 -1j	0.1349	4.6617

OUTPUT SCREENSHOT

```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console

"TEST CASE 2A: VARYING RESISTANCE"

"Resistance R (ohms):"

10.

"Characteristic Impedance Zo (ohms):"

21.002153 - 1.8941759i

"Attenuation Constant alpha (neper/km):"

0.2381132

"Phase Constant beta (radians/km):"

2.6392008

"-----"

"Resistance R (ohms):"

20.

"Characteristic Impedance Zo (ohms):"

21.249174 - 3.7446401i

"Attenuation Constant alpha (neper/km):"

0.4706503

"Phase Constant beta (radians/km):"

2.6702350

"-----"
```

```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console

"-----"

"Resistance R (ohms):"

40.

"Characteristic Impedance Zo (ohms):"

22.119596 - 7.1948864i

"Attenuation Constant alpha (neper/km):"

0.9042246

"Phase Constant beta (radians/km):"

2.7796016

"-----"

"Resistance R (ohms):"

50.

"Characteristic Impedance Zo (ohms):"

22.681319 - 8.7709501i

"Attenuation Constant alpha (neper/km):"

1.1022808

"Phase Constant beta (radians/km):"

2.8501835

"-----"

--> |
```

CODE VARYING INDUCTANCE :

```
clear;

clc;

// Fixed parameters

f = 2.5*10^3;

R = 10;

G = 4*10^-6;

C = 8*10^-6;

w = 2*%pi*f;

// Inductance values in henries

L_values = [3.5 6 8.5 11.5]*10^-3;

disp("TEST CASE 2B: VARYING INDUCTANCE");

for k = 1:4

    L = L_values(k);

    Z = R + %i*w*L;

    Y = G + %i*w*C;

    Zo = sqrt(Z/Y);

    gamma = sqrt(Z*Y);

    disp("Inductance L (mH):");

    disp(L*10^3);

    disp("Characteristic Impedance Zo (ohms):");

    disp(Zo)

    disp("Attenuation Constant alpha (neper/km):");

    disp(real(gamma));
```

```
disp("Phase Constant beta (radians/km):");

disp(imag(gamma));

disp("-----");
```

end

OUTPUT SCREENSHOT

```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console

"TEST CASE 2B: VARYING INDUCTANCE"

"Inductance L (mH):"

3.5

"Characteristic Impedance Zo (ohms):"

21.002153 - 1.8941759i

"Attenuation Constant alpha (neper/km):"

0.2381132

"Phase Constant beta (radians/km):"

2.6392008

"-----"

"Inductance L (mH):"

6.

"Characteristic Impedance Zo (ohms):"

27.424555 - 1.4504082i

"Attenuation Constant alpha (neper/km):"

0.1823734

"Phase Constant beta (radians/km):"

3.4462654

"-----"
```

```
Scilab 6.1.0 Console
File Edit Control Applications ?
Scilab 6.1.0 Console

"-----"

"Inductance L (mH):"

8.5

"Characteristic Impedance Zo (ohms):"

32.618847 - 1.2192897i

"Attenuation Constant alpha (neper/km):"

0.1533509

"Phase Constant beta (radians/km):"

4.0990004

"-----"

"Inductance L (mH):"

11.5

"Characteristic Impedance Zo (ohms):"

37.928904 - 1.0484315i

"Attenuation Constant alpha (neper/km):"

0.1319015

"Phase Constant beta (radians/km):"

4.7662824

"-----"

--> |
```

RESULT – TEST CASE 2

The effect of resistance and inductance on Z_0 and γ was investigated. Increasing resistance mainly increases the attenuation component α , while increasing inductance increases the magnitude of characteristic impedance and phase constant β . The supplied screenshot records one of the changed-parameter executions.

TEST CASE 3 – VARYING FREQUENCY AND CAPACITANCE

The manual asks for frequency variation from 2 kHz to 10 kHz in 2 kHz steps and capacitance variation from 8 μF to 16 μF in 2 μF steps, the remaining constants held fixed. [filecite](#)[turn0file0](#)[L228-L237](#)

CODE VARYING FREQUENCY

```
clear;
clc;

// Fixed parameters
R = 10;
L = 3*10^-3;
G = 4*10^-6;
C = 8*10^-6;

// Frequency values from 2 kHz to 10 kHz
f_values = [2 4 6 8 10]*10^3;

disp("TEST CASE 3A: VARYING FREQUENCY");

for k = 1:5
    f = f_values(k);
    w = 2*pi*f;

    Z = R + %i*w*L;

    Y = G + %i*w*C;

    Zo = sqrt(Z/Y);

    gamma = sqrt(Z*Y);

    disp("Frequency (kHz):");

    disp(f/1000);

    disp("Characteristic Impedance Zo (ohms):");

    disp(Zo);

    disp("Attenuation Constant alpha (neper/km):");

    disp(real(gamma));

    disp("Phase Constant beta (radians/km):");

    disp(imag(gamma));

    disp("-----");
end
```

RELATED TABLE – FREQUENCY VARIATION

f (kHz)	Z_0 (Ω)	α (Np/km)	β (rad/km)
2000	14 -2j	0.3621	2.7769
4000	14 -1j	0.3644	5.5183
6000	14 -1j	0.3648	8.2675
8000	14 -0j	0.3650	11.0187
10000	14 -0j	0.3650	13.7706

RELATED TABLE – CAPACITANCE VARIATION

C (μ F)	Z_0 (Ω)	α (Np/km)	β (rad/km)
8	34 -1j	0.1490	4.2175
10	30 -1j	0.1666	4.7153
12	27 -1j	0.1825	5.1654
14	25 -1j	0.1971	5.5793
16	24 -1j	0.2107	5.9645

OUTPUT SCREENSHOTS

```
Scilab 6.1.0 Console
File Edit Control Applications ?
[Icons]
Scilab 6.1.0 Console

"TEST CASE 3A: VARYING FREQUENCY"

"Frequency (kHz):"

2.

"Characteristic Impedance Zo (ohms):"

19.531674 - 2.5460419i

"Attenuation Constant alpha (neper/km):"

0.2560342

"Phase Constant beta (radians/km):"

1.9635278

"-----"

"Frequency (kHz):"

4.

"Characteristic Impedance Zo (ohms):"

19.407277 - 1.2811807i

"Attenuation Constant alpha (neper/km):"

0.2576743

"Phase Constant beta (radians/km):"

3.9020595

"-----"
```

```
Scilab 6.1.0 Console
File Edit Control Applications ?
[Icons]
Scilab 6.1.0 Console

"-----"

"Frequency (kHz):"

8.

"Characteristic Impedance Zo (ohms):"

19.37555 - 0.6416393i

"Attenuation Constant alpha (neper/km):"

0.2580960

"Phase Constant beta (radians/km):"

7.7913685

"-----"

"Frequency (kHz):"

10.

"Characteristic Impedance Zo (ohms):"

19.371726 - 0.5134128i

"Attenuation Constant alpha (neper/km):"

0.2581469

"Phase Constant beta (radians/km):"

9.7372893

"-----"

--> |
```

CODE VARYING CAPACITANCE :

```
clear;
clc;

// Fixed parameters
R = 10;
L = 9*10^-3;
f = 2.5*10^3;
G = 0.4*10^-6;
w = 2*pi*f;

// Capacitance values from 8 uF to 16 uF
C_values = [8 10 12 14 16]*10^-6;

disp("TEST CASE 3B: VARYING CAPACITANCE");

for k = 1:5
    C = C_values(k);

    Z = R + %i*w*L;

    Y = G + %i*w*C;

    Zo = sqrt(Z/Y);

    gamma = sqrt(Z*Y);

    disp("Capacitance (uF):");

    disp(C*10^6);

    disp("Characteristic Impedance Zo (ohms):");

    disp(Zo);

    disp("Attenuation Constant alpha (neper/km):");

    disp(real(gamma));

    disp("Phase Constant beta (radians/km):");

    disp(imag(gamma));

    disp("-----");
end
```

OUTPUT SCREENSHOT :

```
Scilab 6.1.0 Console
File Edit Control Applications ?
"TEST CASE 3B: VARYING CAPACITANCE"
"Capacitance (uF):"
8.
"Characteristic Impedance Zo (ohms):"
33.561967 - 1.1854772i
"Attenuation Constant alpha (neper/km):"
0.1489849
"Phase Constant beta (radians/km):"
4.2175206
"-----"
"Capacitance (uF):"
10.
"Characteristic Impedance Zo (ohms):"
30.018735 - 1.0603326i
"Attenuation Constant alpha (neper/km):"
0.1665687
"Phase Constant beta (radians/km):"
4.7153315
"-----"
```

```
Scilab 6.1.0 Console
File Edit Control Applications ?
"-----"
"Capacitance (uF):"
14.
"Characteristic Impedance Zo (ohms):"
25.370462 - 0.8961538i
"Attenuation Constant alpha (neper/km):"
0.1970847
"Phase Constant beta (radians/km):"
5.5792555
"-----"
"Capacitance (uF):"
16.
"Characteristic Impedance Zo (ohms):"
23.731894 - 0.8382778i
"Attenuation Constant alpha (neper/km):"
0.2106917
"Phase Constant beta (radians/km):"
5.9644751
"-----"
--> |
```

RESULT – TEST CASE 3

The influence of frequency and capacitance on characteristic impedance and propagation constant was investigated. As frequency increases, β increases strongly, showing the expected increase in phase variation per unit length. Changing capacitance changes both Z_0 and the propagation constant.

OVERALL RESULT

The characteristic impedance (Z_0) and propagation constant ($\gamma = \alpha + j\beta$) of a transmission line were successfully investigated using SCILAB. The three test cases demonstrate the effect of resistance, inductance, frequency and capacitance on the transmission-line secondary parameters. The supplied console screenshots were incorporated as the experimental output evidence, while the related tables present the corresponding calculated values.